DIPLOMA PROGRAMME PRACTICE · HL · SET B

Biology / Paper 1B

Question booklet · 35 marks · 120 minutes across 1A and 1B

Show working, units and suitable precision. Use a calculator. Use separate answer paper for additional working and diagrams.

QUESTION 1 [8]

Testing a proposed clade

Aligned sequences from three ingroup taxa and an outgroup have the six variable sites shown. For this question the outgroup bases are taken as ancestral. Each column represents a different position.

Taxon	1	2	3	4	5	6
Outgroup	A	C	G	T	A	C
R	G	T	G	T	G	C
S	G	T	A	T	A	C
T	A	C	G	C	A	T

(a) Identify two shared derived characters supporting an R–S clade. **[2]**

(b) Calculate the observed R–S and R–T distances as numbers of differing sites. State why these figures are not estimates of genome-wide percentage divergence. **[3]**

(c) A second gene supports an S–T group. Give two biological or sampling reasons for the conflict and an appropriate next step. **[3]**

QUESTION 2 [9]

Filament overlap during contraction

Electron micrographs of comparable human sarcomeres were taken before and after stimulation. All lengths are in μm. The I-band measure in the table is the combined light-region width within one sarcomere.

State	Z-to-Z distance	A band	Combined I regions
Before	2.50	1.60	0.90
After	2.05	1.60	0.45

(a) Calculate percentage shortening of the sarcomere and the percentage change in its combined I-region width. **[3]**

(b) Explain how the data support sliding filaments. **[3]**

(c) A student observes a contracted fibre after ATP supply stops and concludes ATP is unnecessary. Evaluate that inference. **[3]**

QUESTION 3 [9]

Interpreting light and carbon dioxide responses

Net CO_2 uptake was measured on attached leaves under controlled conditions. Each value is the mean of six different plants. Units are µmol CO_2 m^{-2} s^{-1}. Light intensities are in µmol photons m^{-2} s^{-1}.

Light intensity	100	300	600	900
At 400 ppm CO_2	3	8	11	11
At 800 ppm CO_2	3	10	17	19

(a) At light intensity 600 calculate the percentage increase on raising CO_2 and compare it with the change at intensity 100. **[3]**

(b) Interpret the different responses using limiting factors. **[3]**

(c) Explain why an identical pair of means at intensity 100 does not prove a precisely zero effect, and identify two controls or data needed for a stronger comparison. **[3]**

QUESTION 4 [9]

Separating transcription and RNA stability

A hormone is added to cultured cells. Total mature mRNA for gene X and newly synthesized RNA during an equal short pulse are measured relative to untreated samples. After the hormone treatment, transcription is stopped and the fraction of initial mature mRNA remaining is recorded.

Treatment	New RNA in pulse	Mature mRNA at steady state	Fraction remaining after 2 h
No hormone	1.0	1.0	0.50
Hormone	2.0	4.0	0.71

(a) Compare the two RNA measurements before transcription is stopped. Why is transcription alone an incomplete explanation if degradation conditions were unchanged? **[3]**

(b) Assume exponential decay. Estimate the half-life in each treatment using the observation that $0.71^2 \approx 0.50$. **[3]**

(c) Evaluate a claim that the hormone directly binds the promoter of gene X. Give one experiment that distinguishes direct promoter regulation from an indirect response. **[3]**